Method	AppWorld			PlanBench-XL		
	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B
Frozen Base	0.2583 ± 0.0109	0.2720 ± 0.0121	0.3341 ± 0.0101	0.2633 ± 0.0304	0.2682 ± 0.0338	0.3290 ± 0.0429
AREW / T3	0.2771 ± 0.0127	0.2826 ± 0.0087	0.3665 ± 0.0085	0.2809 ± 0.0218	0.3068 ± 0.0433	0.3744 ± 0.0377
GiGPO	0.3094 ± 0.0299	0.3295 ± 0.0268	0.3952 ± 0.0071	0.3163 ± 0.0395	0.3419 ± 0.0558	0.4070 ± 0.0395
InfoPO	0.3634 ± 0.0289	0.3955 ± 0.0035	0.5268 ± 0.0676	0.3975 ± 0.0447	0.4163 ± 0.0449	0.5279 ± 0.0689
WorldEvolver	0.3636 ± 0.0188	0.3827 ± 0.0052	0.4487 ± 0.0081	0.4441 ± 0.0253	0.4541 ± 0.0347	0.5227 ± 0.0228
PoS+EnvProbe	0.5112 ± 0.0021	0.5583 ± 0.0059	0.6649 ± 0.0104	0.4921 ± 0.0234	0.5627 ± 0.0224	0.6630 ± 0.0081
Step-DPO	0.5583 ± 0.0713	0.5871 ± 0.0636	0.6770 ± 0.0398	0.5356 ± 0.0730	0.5607 ± 0.0658	0.6643 ± 0.0358
Ablation-Unweighted	0.5842 ± 0.1161	0.5279 ± 0.1105	0.6484 ± 0.0432	0.5932 ± 0.0605	0.5939 ± 0.0828	0.6386 ± 0.0444
SPAR (Ours)	$\mathbf{0.6130} \pm 0.1138$	$\mathbf{0.6279} \pm 0.1102$	$\mathbf{0.6948} \pm 0.0091$	$\mathbf{0.6243} \pm 0.0470$	$\mathbf{0.6677} \pm 0.0197$	$\mathbf{0.6847} \pm 0.0130$

Table 1: Budget-aware test utility $\mathbb{E}[U_B] \uparrow$ across three backbones (3 seeds, mean $\pm$ standard deviation). AppWorld uses `test_normal` (168 tasks), and PlanBench-XL uses `test` (90 queries). Full metrics are reported in Appendix D.

Metric	RLOO	SPAR	Change
Objective $J(\theta_0)$	0.4640	0.4640	—
Gradient bias	0.0000	**0.0000**	—
Root-gate var. diff.	Baseline	$\mathbf{-0.0164736}$	—
Total grad. variance	0.0711904	**0.0395168**	-44.5%
Pair cost $\mathbb{E}[C]/B$	0.4800	0.5000	—
Variance $\times$ cost (B)	0.0341714	**0.0197584**	-42.2%
Miss rate ($p = 0.02$)	96.04%	**0.00%**	—

Both objectives equal $58/125$. Gradient variance is $\sum_i \mathrm{Var}(\widehat{g}_i)$. Changes are relative to two-sample RLOO where reported.

Table 2: Exact enumeration on the canonical two-stage benchmark ($p = 0.4$ unless specified), verifying Theorems 3.1 and 3.2.

D Complete Cross-Backbone Results

Method	AppWorld				PlanBench-XL			
	$\mathbb{E}[U_B] \uparrow$	$\Delta_{\text{evo}} \uparrow$	$\mathbb{E}[Y](\%) \uparrow$	$\Delta_{\text{use}} \uparrow$	$\mathbb{E}[U_B] \uparrow$	$\Delta_{\text{evo}} \uparrow$	$\mathbb{E}[Y](\%) \uparrow$	$\Delta_{\text{use}} \uparrow$
Frozen Base	0.2583 ± 0.0109	$+0.0000$	34.92	$+0.2042$	0.2633 ± 0.0304	$+0.0000$	35.93	$+0.1167$
AREW / T3	0.2771 ± 0.0127	$+0.0188$	37.70	$+0.2333$	0.2809 ± 0.0218	$+0.0176$	38.52	$+0.1167$
GiGPO	0.3094 ± 0.0299	$+0.0512$	42.46	$+0.2625$	0.3163 ± 0.0395	$+0.0530$	43.70	$+0.1361$
InfoPO	0.3634 ± 0.0289	$+0.1052$	50.40	$+0.3208$	0.3975 ± 0.0447	$+0.1341$	55.56	$+0.2194$
WorldEvolver	0.3636 ± 0.0188	$+0.1053$	50.40	$+0.3694$	0.4441 ± 0.0253	$+0.1807$	62.59	$+0.3208$
PoS+EnvProbe	0.5112 ± 0.0021	$+0.2529$	72.42	$+0.4083$	0.4921 ± 0.0234	$+0.2288$	70.00	$+0.3951$
Step-DPO	0.5583 ± 0.0713	$+0.3000$	79.37	$+0.4958$	0.5356 ± 0.0730	$+0.2723$	75.93	$\mathbf{+0.4142}$
Ablation-Unweighted	0.5842 ± 0.1161	$+0.3260$	**88.49**	$+0.3912$	0.5932 ± 0.0605	$+0.3299$	90.37	$+0.2403$
SPAR (Ours)	$\mathbf{0.6130} \pm 0.1138$	$\mathbf{+0.3547}$	**88.49**	$\mathbf{+0.5033}$	$\mathbf{0.6243} \pm 0.0470$	$\mathbf{+0.3610}$	**91.48**	$+0.3724$

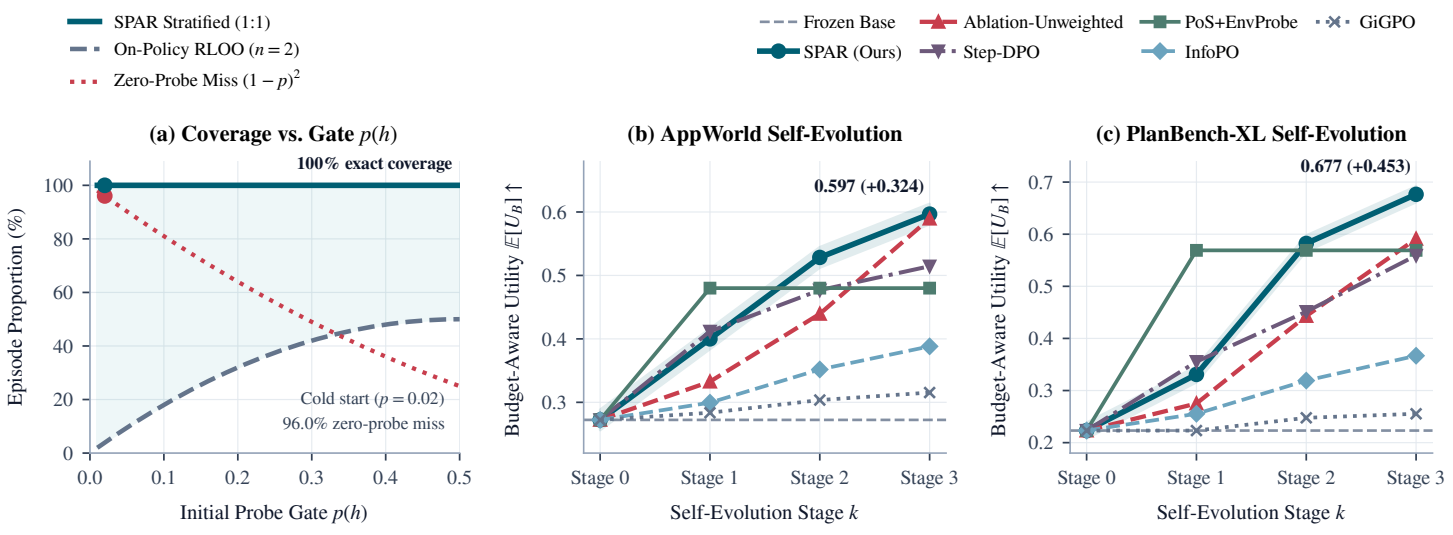
Table 3: Complete results for **Qwen2.5-7B** (3 seeds, mean $\pm$ standard deviation for $\mathbb{E}[U_B]$). AppWorld uses `test_normal` (168 tasks); PlanBench-XL uses `test` (90 queries). Δ_{evo} is the utility gain over Frozen Base, $\mathbb{E}[Y]$ is native terminal success, and Δ_{use} is development-set masked-feedback utilization. Best values within each column are bold.

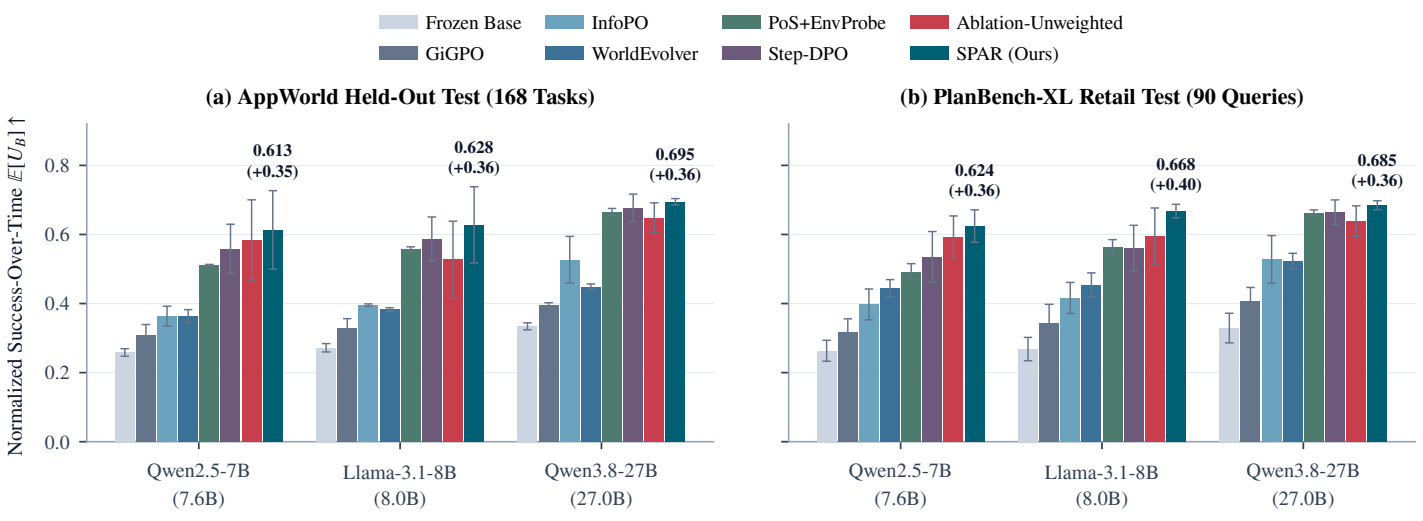
Method	AppWorld				PlanBench-XL			
	$\mathbb{E}[U_B] \uparrow$	$\Delta_{\text{evo}} \uparrow$	$\mathbb{E}[Y](\%) \uparrow$	$\Delta_{\text{use}} \uparrow$	$\mathbb{E}[U_B] \uparrow$	$\Delta_{\text{evo}} \uparrow$	$\mathbb{E}[Y](\%) \uparrow$	$\Delta_{\text{use}} \uparrow$
Frozen Base	0.2720 ± 0.0121	$+0.0000$	36.90	$+0.2333$	0.2682 ± 0.0338	$+0.0000$	36.67	$+0.1167$
AREW / T3	0.2826 ± 0.0087	$+0.0106$	38.49	$+0.2333$	0.3068 ± 0.0433	$+0.0386$	42.22	$+0.1361$
GiGPO	0.3295 ± 0.0268	$+0.0575$	45.44	$+0.2917$	0.3419 ± 0.0558	$+0.0737$	47.41	$+0.1458$
InfoPO	0.3955 ± 0.0035	$+0.1235$	55.16	$+0.3500$	0.4163 ± 0.0449	$+0.1481$	58.52	$+0.2411$
WorldEvolver	0.3827 ± 0.0052	$+0.1107$	53.37	$+0.3694$	0.4541 ± 0.0347	$+0.1859$	64.07	$+0.3403$
PoS+EnvProbe	0.5583 ± 0.0059	$+0.2863$	80.56	$+0.4158$	0.5627 ± 0.0224	$+0.2944$	81.85	$\mathbf{+0.4899}$
Step-DPO	0.5871 ± 0.0636	$+0.3151$	83.53	$+0.5250$	0.5607 ± 0.0658	$+0.2924$	79.63	$+0.4389$
Ablation-Unweighted	0.5279 ± 0.1105	$+0.2559$	77.78	$+0.3897$	0.5939 ± 0.0828	$+0.3256$	90.00	$+0.2586$
SPAR (Ours)	$\mathbf{0.6279} \pm 0.1102$	$\mathbf{+0.3559}$	**90.28**	$\mathbf{+0.5579}$	$\mathbf{0.6677} \pm 0.0197$	$\mathbf{+0.3995}$	**98.15**	$+0.4306$

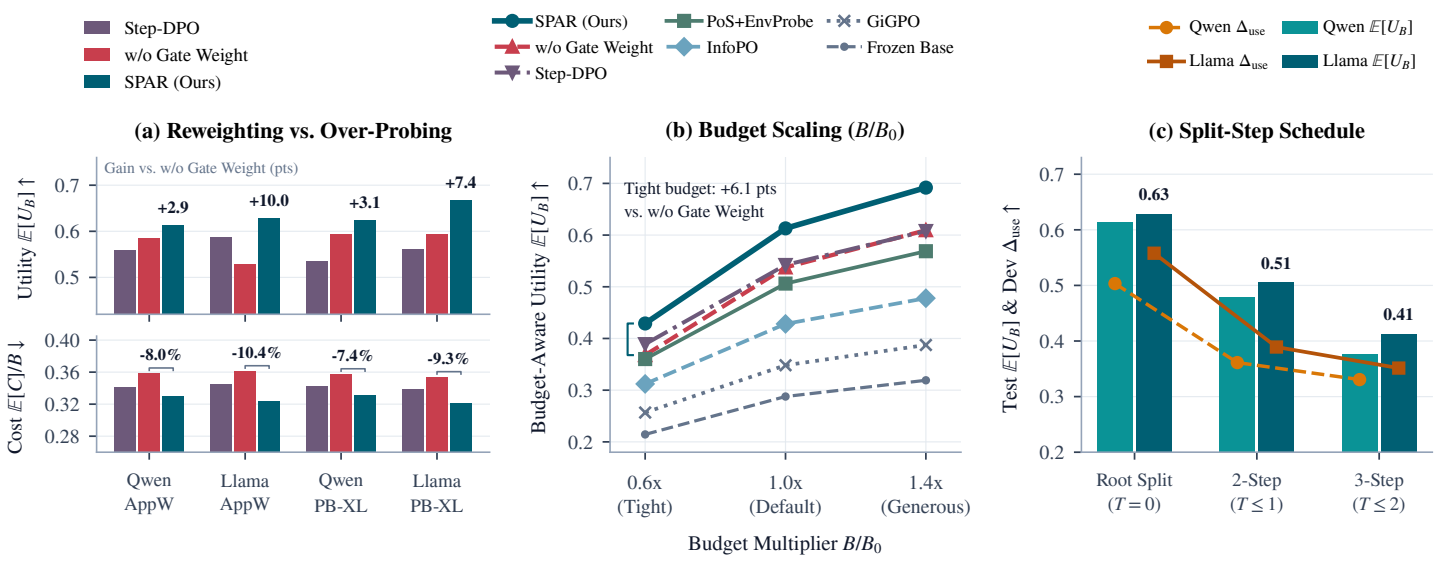
Table 4: Complete results for **Llama-3.1-8B** (3 seeds, mean $\pm$ standard deviation for $\mathbb{E}[U_B]$). AppWorld uses `test_normal` (168 tasks); PlanBench-XL uses `test` (90 queries). Δ_{evo} is the utility gain over Frozen Base, $\mathbb{E}[Y]$ is native terminal success, and Δ_{use} is development-set masked-feedback utilization. Best values within each column are bold.

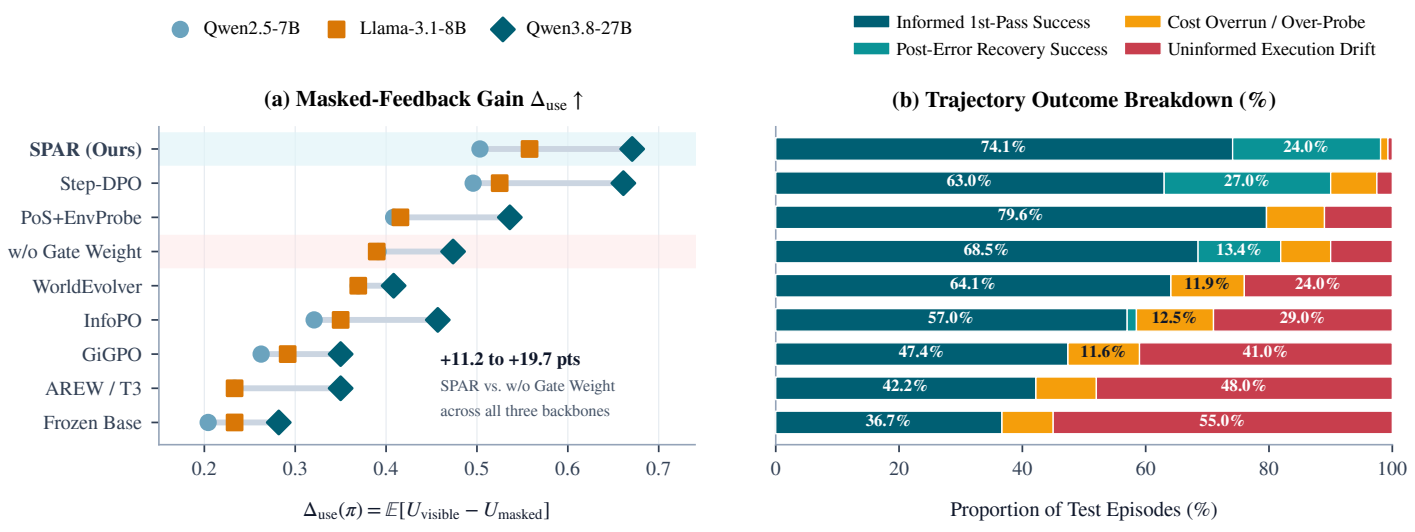
Method	AppWorld				PlanBench-XL			
	$\mathbb{E}[U_B] \uparrow$	$\Delta_{\text{evo}} \uparrow$	$\mathbb{E}[Y](\%) \uparrow$	$\Delta_{\text{use}} \uparrow$	$\mathbb{E}[U_B] \uparrow$	$\Delta_{\text{evo}} \uparrow$	$\mathbb{E}[Y](\%) \uparrow$	$\Delta_{\text{use}} \uparrow$
Frozen Base	0.3341 ± 0.0101	$+0.0000$	46.03	$+0.2819$	0.3290 ± 0.0429	$+0.0000$	45.56	$+0.1458$
AREW / T3	0.3665 ± 0.0085	$+0.0324$	50.79	$+0.3500$	0.3744 ± 0.0377	$+0.0455$	52.22	$+0.1556$
GiGPO	0.3952 ± 0.0071	$+0.0610$	55.16	$+0.3500$	0.4070 ± 0.0395	$+0.0780$	57.04	$+0.2554$
InfoPO	0.5268 ± 0.0676	$+0.1927$	74.60	$+0.4569$	0.5279 ± 0.0689	$+0.1989$	74.81	$+0.4344$
WorldEvolver	0.4487 ± 0.0081	$+0.1145$	63.10	$+0.4083$	0.5227 ± 0.0228	$+0.1937$	74.07	$+0.4861$
PoS+EnvProbe	0.6649 ± 0.0104	$+0.3308$	97.22	$+0.5362$	0.6630 ± 0.0081	$+0.3340$	97.04	$+0.5450$
Step-DPO	0.6770 ± 0.0398	$+0.3429$	96.63	$+0.6611$	0.6643 ± 0.0358	$+0.3353$	94.81	$\mathbf{+0.6021}$
Ablation-Unweighted	0.6484 ± 0.0432	$+0.3142$	96.43	$+0.4737$	0.6386 ± 0.0444	$+0.3096$	94.81	$+0.3746$
SPAR (Ours)	$\mathbf{0.6948} \pm 0.0091$	$\mathbf{+0.3606}$	**99.60**	$\mathbf{+0.6708}$	$\mathbf{0.6847} \pm 0.0130$	$\mathbf{+0.3558}$	**97.78**	$+0.5365$

Table 5: Complete results for **Qwen3.8-27B** (3 seeds, mean $\pm$ standard deviation for $\mathbb{E}[U_B]$). AppWorld uses `test_normal` (168 tasks); PlanBench-XL uses `test` (90 queries). Δ_{evo} is the utility gain over Frozen Base, $\mathbb{E}[Y]$ is native terminal success, and Δ_{use} is development-set masked-feedback utilization. Best values within each column are bold.


SPAR Stratified (1:1)
On-Policy RLOO ($n = 2$)
Zero-Probe Miss $(1-p)^2$
Frozen Base
Ablation-Unweighted
PoS+EnvProbe
GiGPO
SPAR (Ours)
Step-DPO
InfoPO
(a) Coverage vs. Gate $p(h)$
100% exact coverage
Cold start ($p = 0.02$)
96.0% zero-probe miss
Episode Proportion (%)
Initial Probe Gate $p(h)$
(b) AppWorld Self-Evolution
0.597 (+0.324)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
Stage 0
Stage 1
Stage 2
Stage 3
Self-Evolution Stage k
(c) PlanBench-XL Self-Evolution
0.677 (+0.453)
Budget-Aware Utility $\mathbb{E}[U_B] \uparrow$
Stage 0
Stage 1
Stage 2
Stage 3
Self-Evolution Stage k


Frozen Base
InfoPO
PoS+EnvProbe
Ablation-Unweighted
GiGPO
WorldEvolver
Step-DPO
SPAR (Ours)
(a) AppWorld Held-Out Test (168 Tasks)
(b) PlanBench-XL Retail Test (90 Queries)
Normalized Success-Over-Time $\mathbb{E}[U_B] \uparrow$
0.0
0.2
0.4
0.6
0.8
0.613 (+0.35)
0.628 (+0.36)
0.695 (+0.36)
0.624 (+0.36)
0.668 (+0.40)
0.685 (+0.36)
Qwen2.5-7B (7.6B)
Llama-3.1-8B (8.0B)
Qwen3.8-27B (27.0B)


Step-DPO
w/o Gate Weight
SPAR (Ours)
SPAR (Ours)
w/o Gate Weight
Step-DPO
PoS+EnvProbe
InfoPO
GiGPO
Frozen Base
Qwen Δ_{use}
Llama Δ_{use}
Qwen $\mathbb{E}[U_B]$
Llama $\mathbb{E}[U_B]$
(a) Reweighting vs. Over-Probing
Gain vs. w/o Gate Weight (pts)
Utility $\mathbb{E}[U_B]\uparrow$
+2.9
+10.0
+3.1
+7.4
Cost $\mathbb{E}[C]/B\downarrow$
-8.0%
-10.4%
-7.4%
-9.3%
Qwen AppW
Llama AppW
Qwen PB-XL
Llama PB-XL
(b) Budget Scaling (B/B_0)
Tight budget: +6.1 pts
vs. w/o Gate Weight
Budget-Aware Utility $\mathbb{E}[U_B]\uparrow$
0.6x (Tight)
1.0x (Default)
1.4x (Generous)
Budget Multiplier B/B_0
(c) Split-Step Schedule
Test $\mathbb{E}[U_B]$ & Dev $\Delta_{\text{use}}\uparrow$
0.63
0.51
0.41
Root Split ($T=0$)
2-Step ($T\leq 1$)
3-Step ($T\leq 2$)


Qwen2.5-7B
Llama-3.1-8B
Qwen3.8-27B
(a) Masked-Feedback Gain Δ_{use} ↑
SPAR (Ours)
Step-DPO
PoS+EnvProbe
w/o Gate Weight
WorldEvolver
InfoPO
GiGPO
AREW / T3
Frozen Base
0.2
0.3
0.4
0.5
0.6
0.7
+11.2 to +19.7 pts
SPAR vs. w/o Gate Weight
across all three backbones
$\Delta_{use}(\pi) = \mathbb{E}[U_{visible} - U_{masked}]$
Informed 1st-Pass Success
Post-Error Recovery Success
Cost Overrun / Over-Probe
Uninformed Execution Drift
(b) Trajectory Outcome Breakdown (%)
74.1%
24.0%
63.0%
27.0%
79.6%
68.5%
13.4%
64.1%
11.9%
24.0%
57.0%
12.5%
29.0%
47.4%
11.6%
41.0%
42.2%
48.0%
36.7%
55.0%
0
20
40
60
80
100
Proportion of Test Episodes (%)